

Patrick Shannon

Product-Minded Software Architect · Senior Full-Stack Developer

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\$ A / PROFILE

I build systems that stay fast, maintainable, and **useful**.

Senior full-stack developer and architect at **bbox.digital**, taking projects from spec and architecture through implementation, optimization, and deployment — React, Next.js, and TypeScript on the front end; Node.js, Firebase, and GCP behind it. I work closely with the founding team to keep technical decisions aligned with product goals, and standardize AI-assisted workflows across every repo. My differentiator is **provable production outcomes, not capability claims**.

\$ B / PRODUCTION IMPACT

95% ↓

Algolia search requests

100% ↑

CMS API efficiency via
caching

40% ↑

Google Search
impressions

20% ↓

External API calls

60 → 85 LH

Lighthouse mobile score

15+ reL

Releases shipped via
CI/CD

ALGOLIA REQUESTS / MONTH

4,900,000 → 230,000

↓ 95%

Caching + request strategy — product behavior unchanged.

\$ C / EXPERIENCE

Senior Full-Stack Developer

DEC 2021 — PRESENT

bbox.digital · Remote

- Lead architecture and full-stack delivery across several client platforms in parallel — from spec and technical direction through implementation, release coordination, and production hardening.
- Build on React, Next.js, and TypeScript with Node.js, GraphQL/REST APIs, GCP/Firebase, and headless CMS architectures (Sanity, Squidex), including bilingual (FR/EN) routing and Algolia search backends (facets, sort replicas, InstantSearch).
- Modernized Cloud Functions to Gen2 on Node 22 with concurrency tuning for SSR performance; migrated secrets off the deprecated `functions.config()` API to Secret Manager with `dotenv` for local development.
- Hardened production with resilient pipelines — exponential backoff, graceful degradation, per-stage error observability — and a deployment-orchestration layer unifying promote and rollback into a single state machine.
- Standardized AI-assisted workflows across the portfolio: AGENTS.md as a single source of truth per repo, using Claude Code and GitHub Copilot to accelerate architecture, refactoring, and documentation.
- **Outcomes:** +100% CMS API efficiency, -20% external API calls, -95% Algolia search requests (4.9M → 0.23M/mo); CI/CD, Jest/Playwright testing, and SEO/a11y work drove +40% Google Search impressions.

Web Developer & Designer · Webflow Expert

MAY 2021 — DEC 2021

Upwork · Freelance · Remote

- Designed and developed responsive web applications using Figma, Webflow, and JavaScript, with a focus on clean UI/UX, scalability, and maintainability.
- Collaborated with clients in an agile environment, delivering on time with clear communication and iterative product goals.

Porter Airlines · PAL Aerospace · Moncton Flight College

Leadership and training roles in commercial and medevac operations — precision under pressure, team coordination, and regulatory compliance, all directly transferable to software leadership and reliability engineering.

§ D / SELECTED WORK

Classavatar classavatar.ca

Classroom-management platform for Quebec schools — student rewards, avatar progression, parent-teacher messaging, and subscriptions. Full-stack architecture across gamified domain logic and payment infrastructure.

Next.js · React · Firebase · FastAPI · Stripe

Espace OBNL emploi.espaceobnl.ca

Specialized job platform for Quebec nonprofits — authenticated dashboards, marketplace workflows, payments, and search-backed discovery.

Next.js · TypeScript · Firebase · Stripe · Algolia

La Vitrine Culturelle lavitrine.com

Public cultural-event discovery platform — headless content architecture, search performance, custom web components, and serverless integrations.

Webflow · TypeScript · Firebase · GraphQL · Algolia

Tel-jeunes teljeunes.com

Youth-support platform for Quebec — reliable content delivery, structured search, and maintainable backend integration for a sensitive public-facing service.

Webflow · Firebase · GraphQL · Algolia

Captain's Log github.com/patshannon/captains-log

Automated dev journal turning daily GitHub activity into a written narrative — a provider-agnostic LLM pipeline with cost/safety discipline, running unattended nightly via GitHub Actions.

Python · OpenRouter · LLM Pipelines · GitHub Actions

§ E / STACK

FRONTEND	React · Next.js · TypeScript
BACKEND	Node.js · Firebase · Google Cloud Platform
DATA / SEARCH	GraphQL · REST · Algolia · Headless CMS (Sanity, Squidex)
DELIVERY	CI/CD · Jest · Playwright · SEO / a11y · Core Web Vitals
PAYMENTS	Stripe
AI TOOLING	Claude Code · GitHub Copilot · Codex · Spec Kit

§ F / EDUCATION & COMMUNITY

Software Development Training2021 — 2022

Self-Directed

Comprehensive full-stack curriculum — React, Node.js, REST/GraphQL, GCP, system design, testing, and security.

B.Sc., Aviation2009 — 2013

Mount Allison University · Sackville, NB

Graduated with 3.68 GPA.

Mentor · Skills for Hire Atlantic2025 — Ongoing

Digital Nova Scotia · Halifax, NS

Mentoring emerging tech professionals across Atlantic Canada.